

SAO PAULO CONGONHAS, Brazil

WMO: 837800

Lat: 23.627S

Lon: 46.655W

Elev: 2631

StdP: 13.35

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.4	51.4	37.8	36.7	68.2	41.2	42.0	65.6	18.9	69.5	16.7	68.1	6.5	170	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.8	90.0	68.2	88.1	68.2	86.3	68.2	73.7	81.7	72.6	80.8	71.7	80.2	8.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	128.6	77.9	70.1	122.2	76.2	69.5	119.7	75.6	39.1	82.1	38.1	80.8	37.2	80.4	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.1	14.3	DB	44.5	94.1	3.8	1.6	41.7	95.3	39.5	96.2	37.3	97.1	34.5	98.3	(4)
(5)			WB	41.8	76.1	3.6	2.2	39.2	77.7	37.1	79.0	35.1	80.2	32.5	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	74.9	75.7	74.1	71.3	66.2	64.6	64.0	65.9	67.5	69.8	70.7	73.7
	DBStd	6.53	4.33	4.15	4.16	4.57	4.76	5.00	5.63	6.09	6.60	5.96	5.35	4.62
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	361	1	0	0	7	43	67	87	67	50	23	14	2
	CDD50	7240	773	721	746	639	503	439	434	492	525	612	621	735
	CDD65	2126	309	301	282	196	81	56	55	93	125	170	185	273
	CDH74	11340	1629	1739	1366	882	300	213	290	638	874	1025	935	1449
	CDH80	3233	502	558	381	190	30	9	20	138	315	386	271	433
(14) Wind	WSAvg	6.8	6.8	6.6	6.8	6.5	6.2	5.7	5.7	6.4	7.3	7.6	8.0	7.7
(15) Precipitation	PrecAvg	61.5	11.1	8.6	7.7	3.3	2.7	2.4	2.2	1.3	3.6	4.7	6.0	7.9
	PrecMax	86.5	23.8	15.6	15.5	6.4	7.2	10.2	6.3	3.5	8.5	8.0	11.7	14.8
	PrecMin	44.9	6.1	3.5	3.4	0.5	0.6	0.4	0.3	0.1	0.5	0.9	1.9	2.2
	PrecStd	9.6	3.9	2.9	2.6	1.6	1.3	2.1	1.7	0.9	2.2	1.7	2.4	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	91.8	89.8	87.6	82.8	81.0	82.5	87.3	91.5	92.9	89.8	90.7
		MCWB	69.9	70.4	70.0	68.1	65.7	63.8	61.6	62.4	64.1	66.4	67.7	69.1
	2%	DB	88.2	89.3	87.3	84.5	80.4	78.8	79.8	83.9	87.6	88.7	86.3	87.8
		MCWB	70.0	70.0	69.8	67.6	64.5	63.0	60.8	61.4	62.6	66.3	68.1	69.2
	5%	DB	85.7	86.2	84.3	82.3	77.4	76.7	77.3	80.8	84.0	84.4	82.8	84.6
		MCWB	70.3	70.3	69.7	67.6	63.9	61.9	60.6	60.9	63.1	66.8	68.3	69.3
	10%	DB	82.5	83.8	81.4	79.1	74.8	73.7	74.9	77.4	79.0	79.5	79.2	82.1
		MCWB	70.3	70.4	69.6	67.4	63.3	61.2	59.9	60.5	63.2	66.6	67.7	69.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.7	76.0	74.2	72.6	68.8	67.2	65.2	67.5	69.2	72.2	73.7	74.0
		MCDB	82.5	85.0	82.6	80.3	76.4	76.3	75.1	78.1	79.5	81.6	79.7	81.4
	2%	WB	73.1	73.6	72.6	70.9	67.2	65.2	63.5	65.1	67.3	70.0	71.2	72.2
		MCDB	81.2	82.0	81.0	78.4	74.8	73.5	73.4	75.6	78.0	79.8	80.4	80.6
	5%	WB	71.9	72.2	71.4	69.7	65.9	63.7	62.3	63.5	65.8	68.6	69.8	71.1
		MCDB	80.1	80.7	79.8	76.9	73.0	71.8	71.9	73.8	76.0	78.1	78.1	79.6
	10%	WB	71.0	71.2	70.4	68.7	64.8	62.6	61.3	62.1	64.6	67.4	68.6	70.2
		MCDB	78.8	79.5	77.9	75.6	72.0	70.6	70.2	72.0	73.6	76.3	76.5	78.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	14.8	14.1	14.2	13.4	14.7	15.8	17.4	16.3	15.7	14.7	14.8
		MCDBR	18.4	18.7	17.6	17.6	17.2	18.3	20.3	23.0	24.2	23.7	20.1	18.7
		MCWBR	6.8	7.1	6.5	6.5	6.3	6.7	7.2	7.6	7.5	7.9	7.4	7.0
	5% WB	MCDBR	15.7	16.6	15.9	15.1	15.0	16.2	17.5	19.7	19.3	19.4	16.8	16.3
		MCWBR	6.7	7.1	6.8	6.5	6.5	7.0	7.5	8.3	8.0	7.9	7.3	7.0
(40) Clear-Sky Solar Irradiance	taub		0.393	0.392	0.383	0.371	0.349	0.353	0.363	0.397	0.488	0.447	0.416	0.400
	taud		2.469	2.453	2.483	2.489	2.533	2.502	2.459	2.330	2.053	2.240	2.342	2.420
	Ebn at Noon		302	298	292	281	275	266	267	268	256	278	293	300
	Edh at Noon		38	38	35	32	28	28	30	37	53	46	43	40
(44) All-Sky Solar Radiation	RadAvg		1669	1759	1556	1396	1117	1021	1074	1294	1387	1558	1669	1758
	RadStd		237	152	150	115	85	99	114	114	175	180	102	124

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.68	-0.81	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page